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Keeping 90% of ECGs can still lose most AF cases.

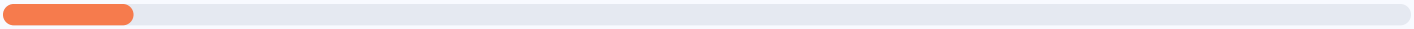
A controlled-noise benchmark of confidence, waveform quality, and selective classification for atrial fibrillation (AF).

01 / THE ORIGINAL FAILURE

At 90% target coverage, the combined filter kept 1,877 of 2,085 test ECGs. Within the 151 AF-positive records, it kept only 14.

All test ECGs 90.02% | 1,877 / 2,085

AF-positive ECGs 9.27% | 14 / 151



Shown: 24 dB recorded mixed artifact, final test set, combined filter. None of the 14 retained AF records was correctly classified.

The primary comparison did not establish superiority.

Combined minus confidence-ranker balanced retained error: -0.26 pp; 95% CI [-2.70, +1.14]. Averaged over four noisy conditions.

02 / A PROMISING CHANGE, TESTED AGAIN

Removing predicted class from the quality ranker improved internal results. A locked followup tested two fresh training seeds and 930 new eligible external ECGs, including 37 AF-positive records.

5 / 5 new comparisons met AF-retention criterion

4 / 5 met the combined improvement + retention rule

Internal gains survived retraining. External improvement remained uncertain for one model, so the complete replication rule was not met.

That external comparison: -1.03 pp balanced error, 95% CI [-4.43, +1.87], versus quality with predicted class. The large internal gains are against that ranker, not against keeping every ECG.

METHOD & SCOPE / 15,292 adult ECGs; 13,796 participants; clinical lead II. Original test: 2,085 ECGs from 1,812 participants. Recorded additive noise, not a wearable trial. The followup reuses the internal test and noise source; external patient linkage is unavailable. The 80% AF-retention criterion is a research rule, not a clinical safety threshold.

Explore the results, manuscript, code, and audit

ksricharan7.github.io/portfolio/ecg.html